

WEEK-8

PART 1: WATER SUPPLY + TREATMENT FLOW

◆ **Complete Water Supply Scheme (FLOW ORDER)**

👉 **Source → Collection → Transport → Treatment → Distribution → Consumer**

★ **MCQ TRAP**

- **Treatment comes BEFORE distribution**
 - **Collection ≠ Transport**
-

◆ **Water Treatment Units (VERY IMPORTANT)**

Impurity Type	Treatment
Suspended particles	Plain sedimentation
Colloidal particles	Coagulation
Dissolved particles	Filtration
Pathogens	Disinfection

★ **Extra Point**

- **Aeration** (ONLY for groundwater) → removes undesirable gases **before sedimentation**
-

◆ **Treatment Sequence (FULL PROCESS)**

1. Pre-treatment → **Coagulation + Flocculation**
2. Main → **Rapid Sand Filtration**
3. Post-treatment → **Chlorination (Disinfection)**

 MCQ Trap

- Filtration does NOT remove pathogens completely → disinfection needed
-

PART 2: PLAIN SEDIMENTATION **(CORE TOPIC)**

◆ **Definition**

- **Solid-liquid separation process**
- Produces:
 - **Overflow** → clarified water (top)
 - **Underflow** → sludge (bottom)

★ **Keywords**

- Overflow = treated water
 - Underflow = sludge
-

◆ **Alternative Name**

- **Sedimentation tank = Settling tank**
-

◆ **Settling Concept**

- Particles settle due to:
 - **Gravity**
 - (or centrifugal force)

PART 3: THEORY OF PARTICLE SETTLING

◆ Forces Acting on Particle

1. Gravitational force (F_g) ↓
2. Buoyant force (F_B) ↑
3. Drag force (F_D) ↑

◆ Drag Force (Stokes Law)

$$F_D = 3\pi\mu D_p v$$

◆ Gravitational Force

$$F_g = \rho_p \frac{\pi}{6} D_p^3 g$$

◆ Buoyant Force

$$F_B = \rho_f \frac{\pi}{6} D_p^3 g$$

◆ Net Downward Force

$$F_{\text{down}} = F_g - F_B - F_D$$

◆ Terminal Velocity (VERY IMPORTANT ★)

👉 When acceleration = 0:

$$v_s = \frac{g(\rho_p - \rho_f)}{18\mu} D_p^2$$

★ Key Conclusions (MCQ GOLD)

- Larger particles → **higher settling velocity**
 - Denser particles → **settle faster**
 - Higher viscosity → **slower settling**
-

PART 4: THEORY OF SEDIMENTATION

♦ **Types of Impurities**

1. **Inorganic** ($SG \approx 2.65$)
2. **Organic** ($SG \approx 1.04$)

★ **MCQ FACT**

- Particles with **$SG \geq 1.2$ settle easily**
-

♦ **Purpose of Settling**

👉 **Remove suspended particles**

♦ **Principle of Settling**

- If **SG of particle $>$ water** → **settles**
 - Requires **reduced turbulence**
 - Tank is called **settling tank**
 - Time spent = **Detention period**
-

PART 5: TYPES OF SETTLING

1. **Type I** → **Discrete settling**
2. **Type II** → **Flocculent settling**

3. Type III → Hindered settling
4. Type IV → Compression settling

◆ Type I (VERY IMPORTANT)

- Particles settle **individually**
 - No interaction
 - **Size, shape, SG constant**
 - **Velocity constant**
-

PART 6: SETTLING BASIN DESIGN

◆ Overflow Rate (Vo) ★

$V_o = \frac{\text{Flow rate}}{\text{Surface area}}$

👉 Surface area = length × width

◆ Horizontal Velocity (VH)

$V_H = \frac{\text{Flow rate}}{\text{Cross-sectional area}}$

👉 Cross-sectional area = height × width

◆ Particle Behavior (VERY IMPORTANT MCQ)

Condition	Result
-----------	--------

$V_s < V_o$	Particle escapes
-------------	------------------

$V_s > V_o$ Particle removed

$V_s = V_o$ Particle just removed

◆ Sedimentation Efficiency ★

$$\eta = \frac{V_s}{V_o} \times 100$$

◆ Design Insight (VERY IMPORTANT)

👉 Larger surface area → better removal efficiency

✓ That's why **wide tanks (Option 2)** are better than deep narrow tanks

ULTRA-FAST REVISION (LAST DAY)

- Treatment order → **Sedimentation** → **Coagulation** → **Filtration** → **Disinfection**
 - Sedimentation removes → **Suspended particles**
 - Stokes law → **velocity** $\propto D^2$
 - Forces → **F_g ↓, F_B ↑, F_D ↑**
 - **$S.G. > \text{water}$** → **settling happens**
 - Efficiency $\propto$ **surface area**
 - **V_s vs V_o decides removal**
-

PRACTICE MCQs

Q1. Which removes pathogens?

- A. Filtration
- B. Sedimentation

- C. Disinfection ☒
 - D. Coagulation
-

Q2. Overflow refers to:

- A. Sludge
 - B. Bottom water
 - C. Top clarified water ☒
 - D. Filtered water
-

Q3. Settling velocity depends on:

- A. Diameter² ☒
 - B. Diameter
 - C. Density only
 - D. Volume
-

Q4. If $V_s < V_o$:

- A. Removed
 - B. Escapes ☒
 - C. Settles fully
 - D. Neutral
-

Q5. Best design for sedimentation:

- A. Deep tank
- B. Small area
- C. Large surface area ☒
- D. High turbulence

PART 2: NUMERICAL ON SEDIMENTATION (VERY IMPORTANT)

◆ Given in Question

- Particle diameter = **0.01 mm**
 - Specific gravity = **2.65**
 - Surface overflow rate (V_o) = **20 m/day**
 - Use **Stokes law**
-

◆ Settling Velocity Formula

$$v_s = \frac{g(\rho_p - \rho_f)}{18\mu} D_p^2 \quad v_s = 18\mu g(\rho_p - \rho_f) D_p^2$$

👉 Final result from slide:

- **$V_s = 7.769 \text{ m/day}$**
-

◆ Efficiency Formula

$$\eta = \frac{V_s}{V_o} \times 100 \quad \eta = \frac{V_s}{V_o} \times 100$$

👉 Final Answer:

- **$\eta = 38.85\%$**
-

MCQ + NUMERICAL TRICKS

- If **$V_s < V_o$** → **efficiency < 100%**
 - Units **MUST** match (convert properly )
 - Diameter in **meters (NOT mm)**
-

PART 3: COAGULATION (VERY IMPORTANT 🔥)

◆ Definition

👉 Used to remove:

- Colloidal particles
 - Fine suspended particles (that escape sedimentation)
-

◆ Steps in Coagulation

1. Addition of coagulant
 2. Flocculation
 3. Clarification (sedimentation)
-

◆ Common Coagulants ★

- Aluminium sulphate (Alum) ★ MOST IMPORTANT
 - Ferric sulphate
 - Ferric chloride
 - Sodium sulphate
-

PART 4: ELECTRICAL DOUBLE LAYER (EDL) ⚡

◆ Concept

- Colloidal particles carry **surface charge (usually negative)**
- Attract **opposite ions (counter-ions)**

◆ Layers

1. Stern layer (fixed)
 2. Diffuse layer (mobile)
-

◆ Important Terms ★

- **Zeta potential** → potential at slipping plane
 - Responsible for **stability of colloids**
-

◆ Key Idea

👉 EDL creates **repulsion between particles** → prevents settling

PART 5: DESTABILIZATION OF COLLOIDS

◆ Methods (VERY IMPORTANT MCQ)

1. Charge neutralization
 2. Sweep coagulation
 3. Bridging
 4. Patch flocculation
-

◆ Final Result

👉 Formation of **flocs (larger particles)** → easier to settle

PART 6: JAR TEST (IMPORTANT PRACTICAL MCQ)

👉 Used to:

- Find **optimum coagulant dose**

♦ Graph Concept

- X-axis → **Coagulant dose**
- Y-axis → **Turbidity**

👉 Curve shows:

- Minimum turbidity → **optimum dose**

♦ Zones (VERY IMPORTANT ⚠)

Zone	Meaning
------	---------

Zone 1	Insufficient dose
--------	-------------------

Zone 2	Proper destabilization
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Zone 3	Restabilization (overdose)
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Zone 4	Sweep flocculation
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PART 7: FILTRATION

◆ Goal

👉 Reduce turbidity from:

- ~3 TU → below 0.3 TU
-

◆ Types of Filters

1. Slow sand filter
 2. Rapid sand filter
 3. Pressure filter
-

◆ Mechanisms of Filtration ★

- Mechanical straining
 - Sedimentation + adsorption
 - Biological metabolism
 - Electrolytic action
-



PART 8: SLOW SAND FILTER (VERY IMPORTANT 🔥)

◆ Key Features

- Uses sand + gravel
- Gravity driven
- No chemicals added
- Biofilm layer (IMPORTANT)

◆ Biofilm Role ★

👉 Removes:

- Bacteria
- Viruses
- Protozoa
- Colloids

◆ History (FACT BASED MCQ)

- First used in **1790 (England)**
- Widely used by **1800s**
- WHO → calls it **efficient + economical**

◆ Advantages ★

- Long life
- Low cost
- Low maintenance
- Minimal sludge
- No skilled operator needed
- Low power & chemicals

◆ Best For (VERY IMPORTANT)

👉 Rural areas

◆ Why Suitable for Rural Areas?

- Simple technology
 - Low cost
 - Good pathogen removal
 - Low energy
 - Minimal supervision
-

PART 9: COMPARISON (SLOW vs RAPID FILTER)

◆ Key Differences

Feature	Slow Sand	Rapid Sand
Pre-treatment	Not needed	Needed
Filtration rate	Low	High
Efficiency	High	Moderate
Cost	Low	High
Maintenance	Low	High

ULTRA-FAST REVISION

- Coagulation removes → **colloids**
- EDL → **causes stability**
- Zeta potential → **measure of stability**
- Jar test → **optimum dose**
- Zone 3 → **overdose**
- Slow sand filter → **biofilm based**

- Rural → **slow sand best**
-

PRACTICE MCQs

Q1. Zeta potential indicates:

- A. Velocity
 - B. Stability of colloids ☒
 - C. Size
 - D. Density
-

Q2. Best coagulant:

- A. NaCl
 - B. Alum ☒
 - C. KCl
 - D. CaCO_3
-

Q3. Jar test is used for:

- A. Filtration
 - B. Sedimentation
 - C. Coagulant dose ☒
 - D. Disinfection
-

Q4. Zone 3 represents:

- A. Ideal
 - B. Underdose
 - C. Overdose ☒
 - D. Neutral
-

Q5. Slow sand filter works by:

- A. Pressure
 - B. Chemicals
 - C. Biofilm + gravity ☒
 - D. Heat
-

Q6. Which layer is fixed in EDL?

- A. Diffuse
- B. Stern layer ☒
- C. Outer layer
- D. Slipping layer

📖 PART 3: RAPID SAND FILTER (RSF) 🔥

♦ Basic Structure

Layers:

Sand

Gravel (support layer)

Components:

Inlet chamber

Wash water troughs

Underdrain system

Outlet (filtered water)

♦ Filtration Process

👉 Water flows:

From top → bottom

Through sand → impurities removed

♦ Backwashing Process (VERY IMPORTANT ★)

👉 Reverse flow (bottom → top)

Purpose:

Remove trapped particles

Clean filter bed

♦ Key MCQ Points

Backwashing uses clean water

Done periodically (≈ every 2 days)

Requires pumps (power needed)

▣ PART 4: RAPID SAND FILTER (THEORY)

◆ Key Characteristics

Used in developed countries

Treats large quantity of water

Requires:

Pre-treatment (coagulation + sedimentation) ★

Pumps for backwashing

◆ Important Facts ⚠

Produces sludge + wash water

Needs treatment before disposal

Land required = ~20% of slow sand filter ★

▣ PART 5: OPERATION & MAINTENANCE

◆ Key Points

Regular backwashing required

Removes deposited suspended matter

May need repacking of filter bed

Requires trained personnel ★

▣ PART 6: COST

◆ Major Cost Components

Materials (cement, sand, gravel, steel)

Pipes & valves

Energy (IMPORTANT ⚠)

👉 Labor cost = less significant

▣ PART 7: EFFECTIVENESS & SUITABILITY

- ◆ Effectiveness

Very effective for suspended solids

BUT:

Needs pre-treatment

Disinfection required after filtration

- ◆ Best Suitable For ★

Urban areas

Large-scale water supply

Areas with limited land

▣ PART 8: ADVANTAGES & DISADVANTAGES

- ◆ Advantages

High efficiency

Requires less land

Proven technology

- ◆ Disadvantages ⚠

High cost

Needs energy

Needs skilled operator

Requires pre-treatment

▣ PART 9: URBAN vs RURAL (VERY IMPORTANT MCQ)

- ◆ Urban Areas:

High population density

High land cost

Limited space

👉 Use Rapid Sand Filter

♦ Rural Areas:

👉 Use Slow Sand Filter

Low cost

Simple

No skilled labor

📖 PART 10: NUMERICALS (VERY IMPORTANT 🔥🔥)

♦ 1. Slow Sand Filter Design

Formula:

$A = Q \text{ rate of filtration}$
 $A = \text{rate of filtration} \times Q$

👉 Given:

$Q = 2 \text{ MLD}$

Rate = 200 L/hr/m²

👉 Final:

Area = 416.66 m²

Width = 14.43 m

Length = 28.86 m

★ Trick

Always convert:

MLD → L/hr or m³/day

♦ 2. Rapid Sand Filter Design

👉 Given:

$Q = 30 \text{ MLD}$

$\text{Rate} = 3000 \text{ L/hr/m}^2$

$\text{Filters} = 5$

Step 1: Total Area

416.66 m^2

Step 2: Area per filter

83.32 m^2

Step 3: Dimensions

$\text{Width} = 6.45 \text{ m}$

$\text{Length} = 12.9 \text{ m}$

◆ 3. Backwash Water Calculation ★

Formula:

$\text{Volume} = \text{Rate} \times \text{Time} \times \text{Area}$

👉 Result:

$244,960 \text{ m}^3$

◆ 4. % Water Used in Backwash ★

$\% = \frac{\text{Backwash water}}{\text{Filtered water}} \times 100\% = \frac{\text{Filtered water}}{\text{Backwash water}} \times 100$

👉 Final Answer:

$\approx 4.2\%$

⚠ MCQ TRAPS

$\text{Backwash rate} = 6 \times \text{filtration rate}$

Time must be in hours

Use area of ONE filter

ULTRA-FAST REVISION (FINAL)

RSF → requires pre-treatment

Backwashing → reverse flow

Done every ~2 days

RSF → urban areas

SSF → rural areas

RSF land = 20% of SSF

Backwash water $\approx$ 4–5%

PRACTICE MCQs

Q1. Backwashing is:

- A. Forward flow
- B. Reverse flow 
- C. Circular flow
- D. No flow

Q2. RSF requires:

- A. No treatment
- B. Pre-treatment 
- C. Heating
- D. Cooling

Q3. Suitable for urban areas:

- A. Slow sand
- B. Rapid sand 
- C. Both
- D. None

Q4. Backwash frequency:

- A. Daily
- B. Weekly

- C. Every ~2 days ☒
 - D. Monthly
-

Q5. Land required by RSF:

- A. More
 - B. Equal
 - C. Less ($\approx 20\%$) ☒
 - D. None
-

Q6. Backwash water %:

- A. 1%
- B. 2%
- C. 4–5% ☒
- D. 10%



PART 4: DISINFECTION

◆ Definition

👉 Killing of pathogenic bacteria

◆ Methods of Disinfection

- Boiling
 - Excess lime treatment
 - **Ozone**
 - Iodine & bromine
 - **UV rays**
 - KMnO_4
 - Silver (electro-katadyn)
 - **Chlorination** ★ (MOST IMPORTANT)
-



DESIRED PROPERTIES OF DISINFECTANTS ★

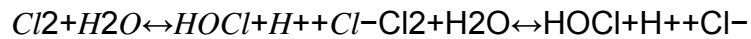
1. Kill pathogens **quickly**
 2. Work under **varying conditions**
 3. **Non-toxic & palatable**
 4. Easy to **store & use**
 5. Easy to **measure concentration**
 6. Provide **residual effect** ★
 7. **Cheap**
-

PART 5: CHLORINATION (VERY IMPORTANT 🔥🔥)

◆ Chlorine Forms

- Cl_2
 - NaOCl
 - Ca(OCl)_2
-

◆ Reaction in Water



◆ pH Dependence ★

Form	Condition
HOCl (strong disinfectant)	pH < 6 ★
OCl^-	pH > 7.5
Cl_2	pH < 1

◆ Free Available Chlorine ★

👉 Chlorine present as:

- $\text{HOCl} + \text{OCl}^-$
-

PART 6: STRENGTH OF DISINFECTANTS

👉 Order:

$\text{O}_3 > \text{ClO}_2 > \text{Free Chlorine} > \text{Chloramines}$

◆ Key Points

- O_3 & $\text{ClO}_2 \rightarrow$ **no residual**
 - Free chlorine $\rightarrow$ **cheap**
 - Chloramines:
 - Less reactive
 - **Longer residual**
-

PART 7: COMBINED CHLORINE

◆ Reactions

- $\text{NH}_3 + \text{HOCl} \rightarrow \text{NH}_2\text{Cl}$ (**Monochloramine**)
 - $\text{NH}_2\text{Cl} + \text{HOCl} \rightarrow \text{NHCl}_2$ (**Dichloramine**)
 - $\text{NHCl}_2 + \text{HOCl} \rightarrow \text{NCl}_3$ (**Trichloramine**)
-

◆ Important Fact

👉 Combined chlorine is **~25× less effective than free chlorine**



PART 8: BREAKPOINT CHLORINATION



◆ Concept

- As chlorine dose increases:
 - First → forms chloramines
 - Then → destroys them
 - Finally → **free chlorine appears**

👉 This point = **BREAKPOINT**

◆ MCQ TRICK

- After breakpoint → **free chlorine residual increases sharply**
-



PART 9: NUMERICAL (CHLORINE DEMAND)

◆ Formula

Chlorine demand = Dose - Residual

👉 Given:

- Dose = 0.6 mg/L
- Residual = 0.2 mg/L

👉 Demand = **0.4 mg/L**

◆ Bleaching Powder Requirement

Required=Dose%available chlorine
Required=%available chlorineDose

👉 Result:

- 2 mg/L
-

◆ Monthly Requirement ★

- 600 kg/month
-

⚠ MCQ TRAPS

- % chlorine matters
 - Convert MLD properly
 - Use **dose**, not **demand** in some questions (careful!)
-

PART 10: WATER SUPPLY SYSTEM

◆ Purpose

👉 Meet demands:

- Domestic
- Commercial
- Industrial
- Fire-fighting

👉 Must provide:

- **Adequate pressure**
-



COMPONENTS

- Pipe system
- Pumping stations
- Storage (reservoirs)
- Fire hydrants
- House connections
- Meters



PART 11: SYSTEM CONFIGURATIONS

◆ Types

1. **Branching system**
2. **Grid system** 
3. Combination

◆ Comparison

Feature	Branchin g	Grid
Supply direction	One	Multiple 
Reliability	Low	High
Preferred		

◆ Key Insight

👉 **Grid system preferred** (better distribution)

PART 12: BASIC REQUIREMENTS

◆ Pressure

- Must be **sufficient**
 - Not too high (cost + leakage)
-

◆ Capacity

- Based on:
 - Water demand
 - Fire demand
-

◆ Pipe Size ★

- Based on **flow velocity**
 - Minimum **6 inches for fire demand**
-

PART 13: HYDRAULIC DESIGN

◆ Design Flow Rate

👉 Maximum of:

- Max daily demand + fire demand
 - Max hourly demand
-

◆ **Skeletonization** ★

👉 Replace complex network with:

- **Equivalent pipe system**
-

PART 14: DISTRIBUTION RESERVOIRS

◆ **Functions** ★

- Provide **storage**
 - Maintain **pressure (head)**
 - Meet:
 - Fire demand (10–12 hrs)
 - Emergency demand
 - Reduce friction losses
-

◆ **Location**

- Near **center of use**
-

PART 15: PUMPING SYSTEM

◆ **Types of Pumps**

- Centrifugal ★
- Displacement

◆ Types of Pumping

- Low lift → source to plant
- High service → to distribution
- Booster → increase pressure
- Recirculation → within plant
- Well pumps

◆ System Types ★

- Gravity system
- Pumping system
- Dual system

🧠 FINAL ULTRA-REVISION (MOST IMPORTANT 🔥)

- Best disinfectant → **Chlorine**
- Strongest → **Ozone**
- HOCl works best at **low pH**
- Breakpoint → **free chlorine appears**
- Grid system → **best distribution**
- RSF → **urban**, SSF → **rural**
- Reservoir → **storage + pressure**
- Pump types → **centrifugal**

🎯 FINAL PRACTICE MCQs

Q1. Best disinfectant:

- A. UV
 - B. Chlorine ☒
 - C. KMnO_4
 - D. Lime
-

Q2. Strongest disinfectant:

- A. Chlorine
 - B. Ozone ☒
 - C. Chloramine
 - D. UV
-

Q3. Breakpoint indicates:

- A. No chlorine
 - B. Combined chlorine
 - C. Free chlorine appears ☒
 - D. End of reaction
-

Q4. Best distribution system:

- A. Branch
 - B. Grid ☒
 - C. Circular
 - D. None
-

Q5. Reservoir purpose:

- A. Cooling
 - B. Storage + pressure ☒
 - C. Filtration
 - D. Pumping
-

Q6. Most used pump:

- A. Gear
- B. Reciprocating
- C. Centrifugal ☒
- D. Jet

1. WATER TREATMENT FLOW

 **Source** → **Collection** → **Transport** → **Treatment** → **Distribution** → **Consumer**

2. TREATMENT vs IMPURITY

Impurity	Treatment
----------	-----------

Suspended	Sedimentation
-----------	---------------

Colloidal	Coagulation
-----------	-------------

Dissolved	Filtration
-----------	------------

Pathogens	Disinfection
-----------	--------------

3. SEDIMENTATION

✓ Key Points

- Removes suspended particles
- Output:
 - Overflow → clean water

- Underflow → sludge
-

✓ Forces on Particle

- Gravity ↓
 - Buoyancy ↑
 - Drag ↑
-

✓ Settling Velocity (IMPORTANT)

$$v_s = \frac{g(\rho_p - \rho_f)}{18\mu} D_p^2 \quad v_s = 18\mu g(\rho_p - \rho_f) D_p^2$$

👉 Depends on:

- Particle size (D^2)
 - Density
 - Viscosity
-

✓ Conditions ★

- $V_s > V_o \rightarrow$ **Removed**
 - $V_s < V_o \rightarrow$ **Escapes**
-

✓ Efficiency

$$\eta = \frac{V_s}{V_o} \times 100 \quad \eta = \frac{V_o}{V_s} \times 100$$

✓ Types

- Type I → **Discrete (MOST IMPORTANT)**
- Type II → Flocculent
- Type III → Hindered
- Type IV → Compression

4. COAGULATION

✓ Purpose

👉 Remove **colloids**

✓ Steps

1. Coagulant addition
 2. Flocculation
 3. Sedimentation
-

✓ Coagulants ★

- Alum (**MOST IMPORTANT**)
 - Ferric salts
-

✓ EDL (Electrical Double Layer)

- Causes **repulsion** → **stability**
 - **Zeta potential** = **stability measure**
-

✓ Destabilization Methods

- Charge neutralization
 - Sweep coagulation
 - Bridging
 - Patch flocculation
-

✓ Jar Test ★

👉 Finds **optimum dose**

Zone	Meaning
-------------	----------------

1	Underdose
---	-----------

2	Ideal
---	-------

3	Overdose ⚠️
---	----------------

4	Sweep
---	-------

5. FILTRATION

✓ Types

- Slow sand filter
 - Rapid sand filter ★
 - Pressure filter
-

✓ Mechanisms

- Straining
 - Adsorption
 - Biological action
 - Electrolytic action
-

6. SLOW vs RAPID FILTER

Feature	Slow	Rapid
Cost	Low	High
Land	High	Low
Skill	No	Yes
Pre-treatment	No	Yes
Area	Rural	Urban

7. RAPID SAND FILTER

✓ Key Points

- Requires **pre-treatment**
 - Uses **backwashing**
-

✓ Backwashing

- Reverse flow
- Done every **~2 days**

- Removes dirt
-

✓ Important

- Needs **power + operator**
 - Land $\approx$ **20% of slow filter**
-

8. DISINFECTION

✓ Purpose

 Kill pathogens

✓ Methods

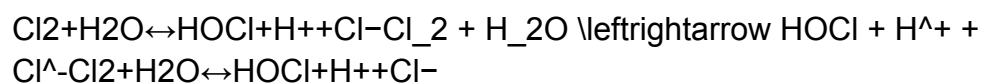
- Chlorine 
 - Ozone
 - UV
 - KMnO_4
-

✓ Best Disinfectant

 Chlorine

9. CHLORINATION

✓ Reaction



✓ pH Effect

- Low pH → **HOCl (best)**
- High pH → OCl^-

✓ Strength Order

👉 $\text{O}_3 > \text{ClO}_2 > \text{Cl}_2 > \text{Chloramines}$

✓ Breakpoint ★

👉 Point where:

- Chloramines destroyed
- **Free chlorine appears**

✓ Chlorine Demand

$\text{Demand} = \text{Dose} - \text{Residual}$
 $\text{Demand} = \text{Dose} - \text{Residual}$

📌 10. WATER SUPPLY SYSTEM

✓ Demands

- Domestic
- Industrial
- Fire

✓ Components

- Pipes
 - Pumps
 - Reservoirs
 - Hydrants
-

11. DISTRIBUTION SYSTEM

✓ Types

- Branching
 - **Grid (BEST)**
-

✓ Key Fact

👉 Grid → supply from **multiple directions**

12. RESERVOIRS

✓ Functions

- Storage
 - Pressure
 - Fire demand
 - Emergency
-

✓ Location

👉 Near **center**

13. PUMPING

✓ Types

- Centrifugal (MOST USED)
-

✓ Systems

- Gravity
 - Pumping
 - Dual
-

14. NUMERICAL QUICK FORMULAS

✓ Area

$$A = \frac{Q}{\text{rate}} \quad A = \frac{Q}{\text{rate}}$$

✓ Efficiency

$$\eta = \frac{V_s}{V_o} \times 100 \quad \eta = \frac{V_o}{V_s} \times 100$$

✓ Chlorine Demand

$$\text{Chlorine Demand} = \text{Dose} - \text{Residual}$$

MOST COMMON MCQ TRAPS

- Sedimentation $\neq$ coagulation
- Filtration $\neq$ disinfection
- Chloramine < chlorine strength
- RSF needs **pre-treatment**
- Backwashing = **reverse flow**
- Grid > Branching

- HOCl best at **low pH**
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LAST 5-MIN REVISION

👉 Just remember:

- **Flow: Source → Treatment → Distribution**
- **Sedimentation → Suspended**
- **Coagulation → Colloids**
- **Filtration → Fine particles**
- **Disinfection → Pathogens**
- **Chlorine = best**
- **Grid system = best**
- **RSF = urban, SSF = rura**